Decision Tree Classification

Classification example • Jupyter Notebook-style code and output

In [1]:

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [2]:

```
X, y = make_classification(n_samples=180, n_features=2,
                           n_redundant=0, n_informative=2,
                           n_clusters_per_class=1,
                           class_sep=1.35, random_state=7)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)
```

Out [ ]:

```
Training samples: 135
Test samples: 45
```

In [3]:

```
# Train a decision tree classifier
model = DecisionTreeClassifier(max_depth=3, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

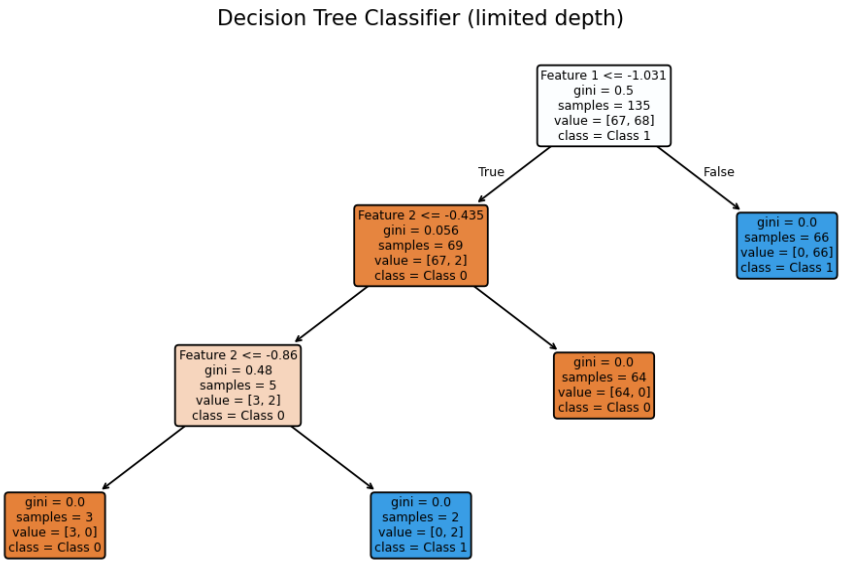
print("Accuracy:", round(accuracy_score(y_test, y_pred), 3))
print("Confusion matrix:\n", confusion_matrix(y_test, y_pred))
```

Out [ ]:

```
Accuracy: 0.978
Confusion matrix:
[[23  0]
 [ 1 21]]
```

In [4]:

```
# Plot the trained decision tree
plt.figure(figsize=(12, 6))
plot_tree(model, filled=True,
          feature_names=['Feature 1', 'Feature 2'],
          class_names=['Class 0', 'Class 1'],
          rounded=True)
plt.title('Decision Tree Classifier')
plt.show()
```



Rendered output of the plotting cell